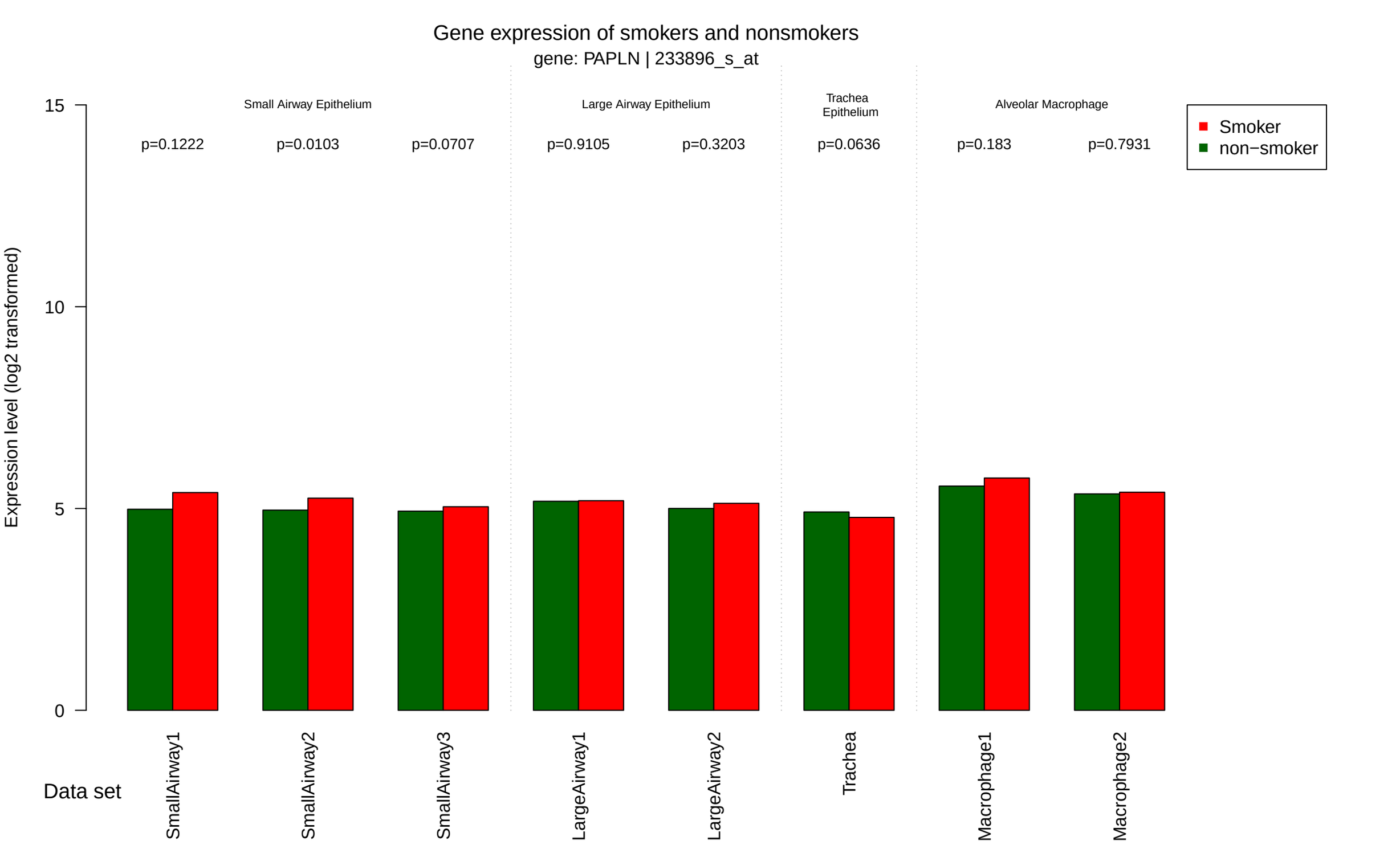
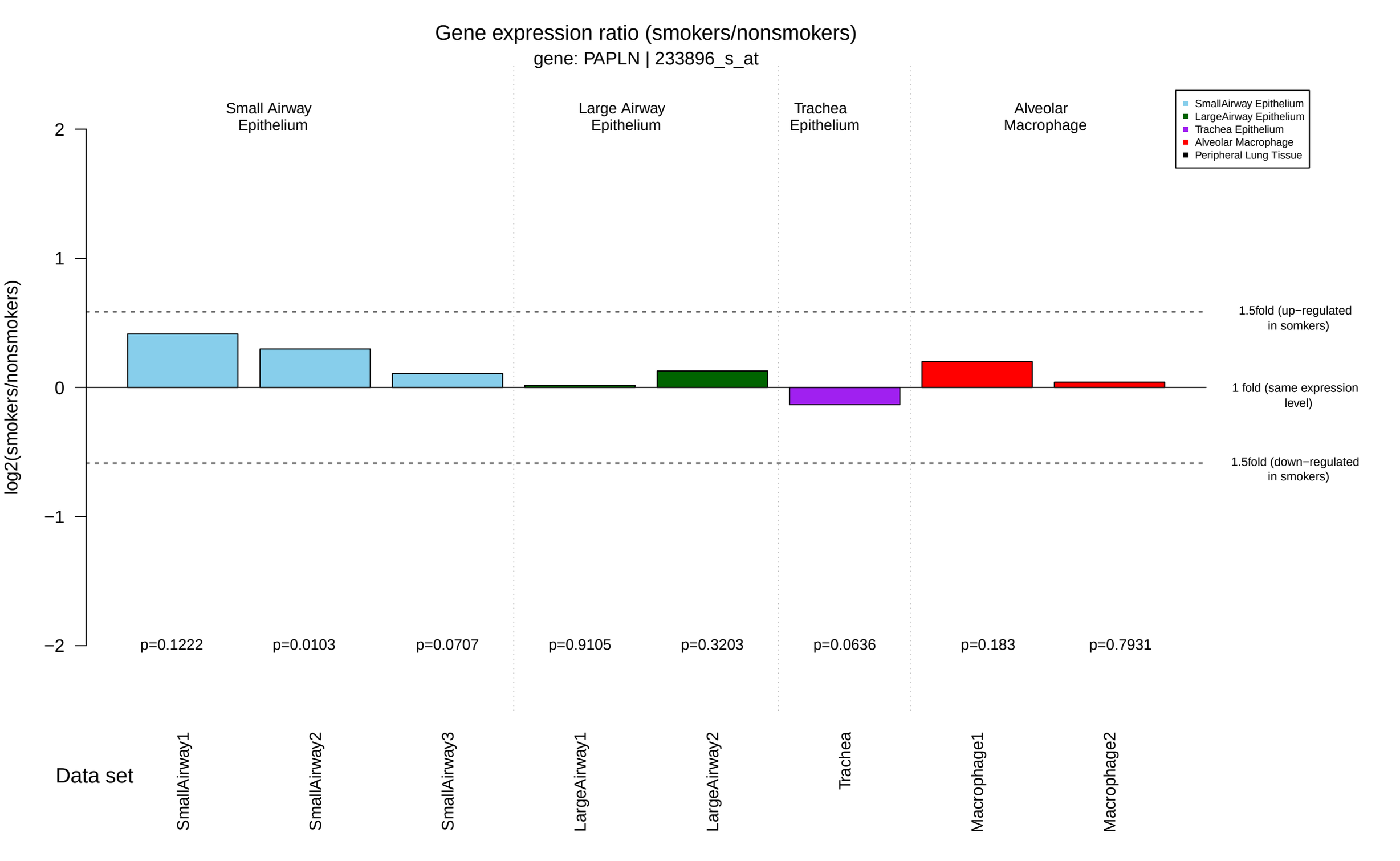
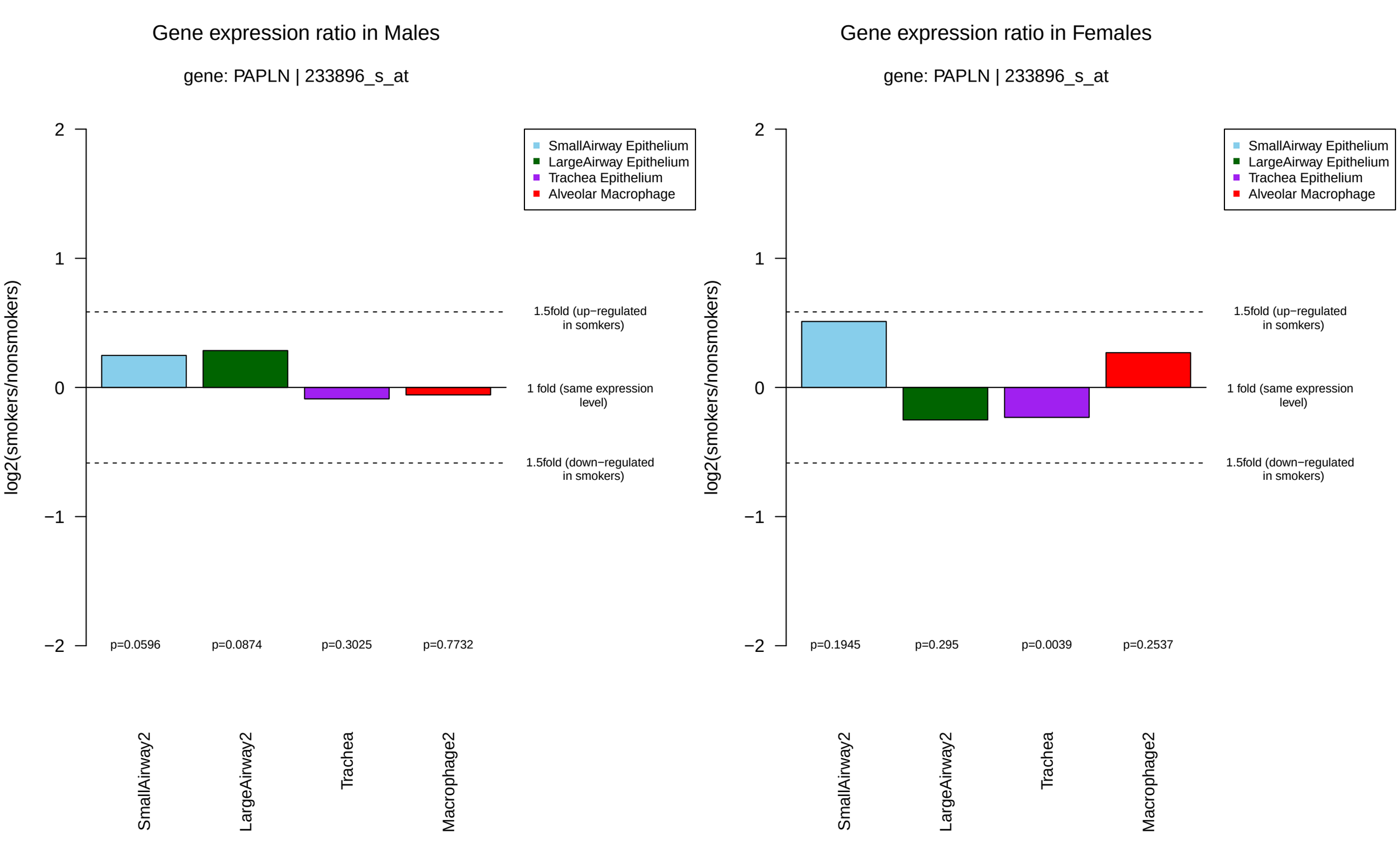
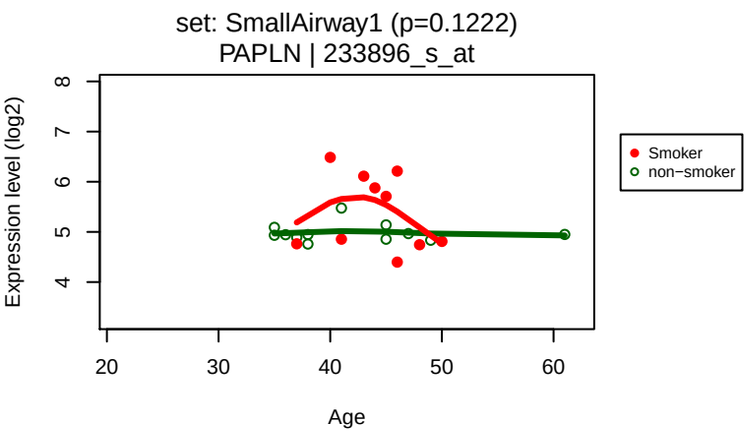
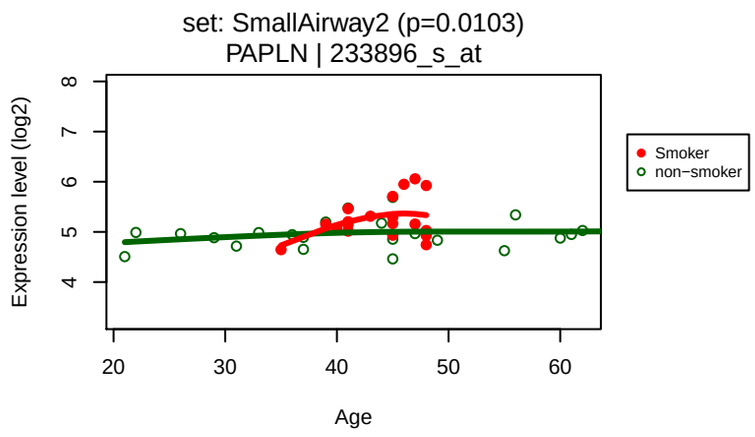
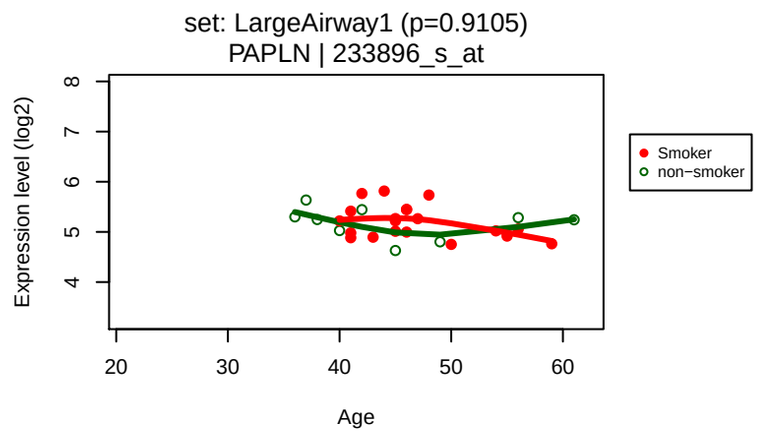

Gene expression of smokers and nonsmokers
gene: PAPLN | 233896_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1222
p=0.0103
p=0.0707
p=0.9105
p=0.3203
p=0.0636
p=0.183
p=0.7931
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

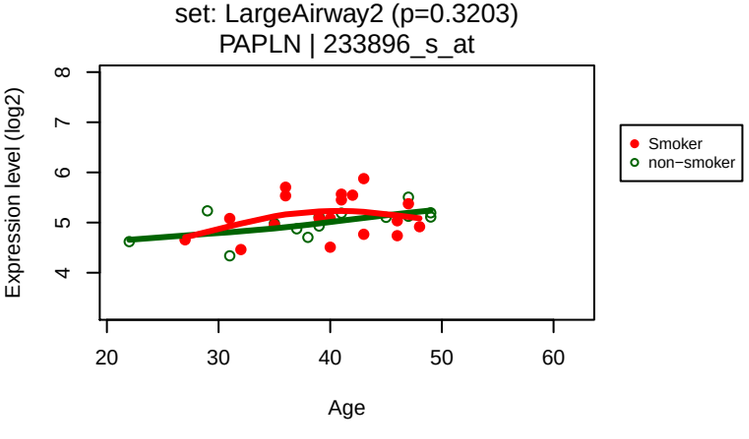

Gene expression ratio (smokers/nonsmokers)
gene: PAPLN | 233896_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1222
p=0.0103
p=0.0707
p=0.9105
p=0.3203
p=0.0636
p=0.183
p=0.7931
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

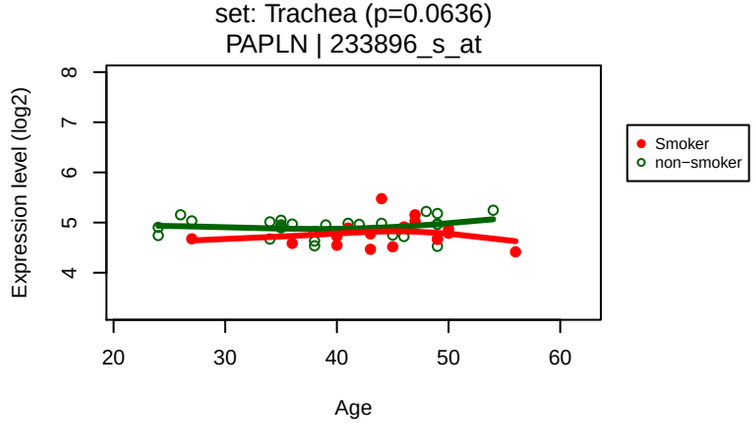

Gene expression ratio in Males
gene: PAPLN | 233896_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0596
p=0.0874
p=0.3025
p=0.7732
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PAPLN | 233896_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1945
p=0.295
p=0.0039
p=0.2537
SmallAirway2
LargeAirway2
Trachea
Macrophage2

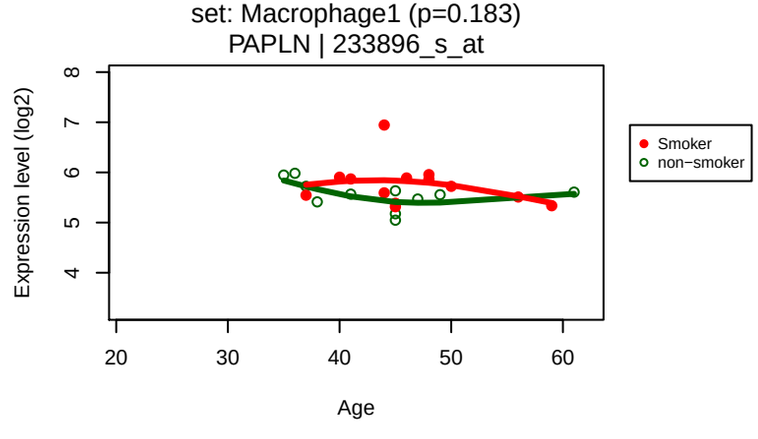

set: SmallAirway1 (p=0.1222)
PAPLN | 233896_s_at
Expression level (log2)
Age
Smoker
non−smoker

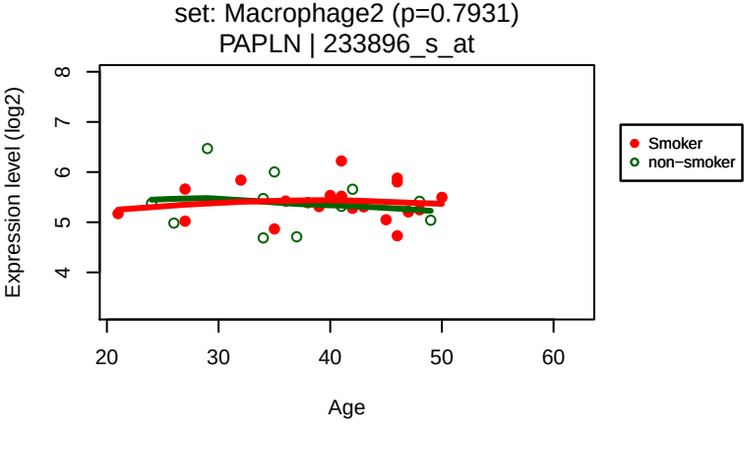

set: SmallAirway2 (p=0.0103)
PAPLN | 233896_s_at
Expression level (log2)
Age
Smoker
non−smoker

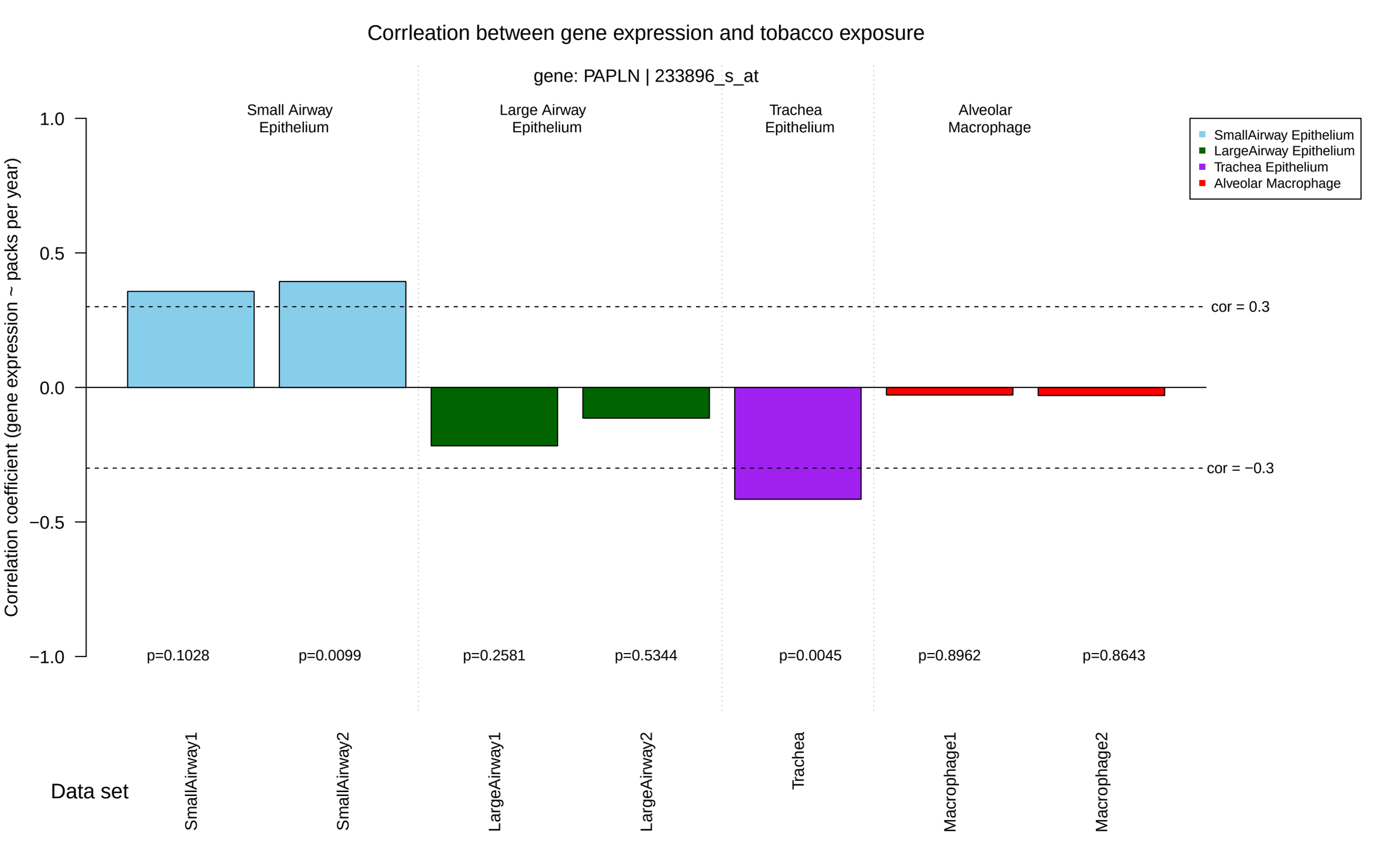

set: LargeAirway1 (p=0.9105)
PAPLN | 233896_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3203)
PAPLN | 233896_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0636)
PAPLN | 233896_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.183)
PAPLN | 233896_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7931)
PAPLN | 233896_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PAPLN | 233896_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1028
p=0.0099
p=0.2581
p=0.5344
p=0.0045
p=0.8962
p=0.8643
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2